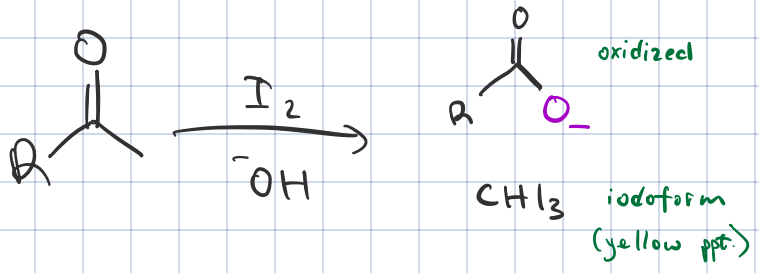
Iodoform Test

Tests for Me–Ketones:

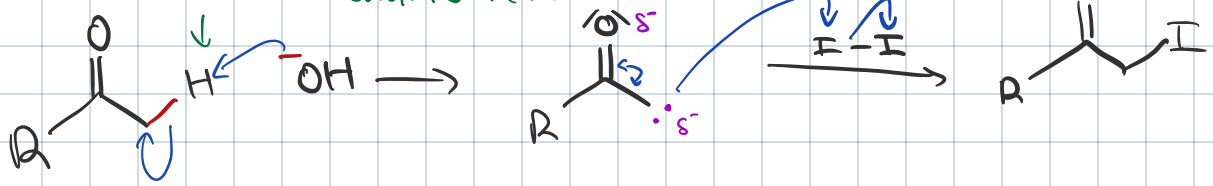


Haloform Mech

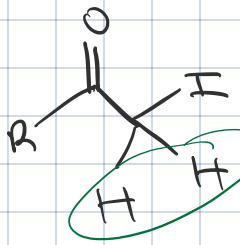
acidic → enolate resonance

α-halogenation

1.

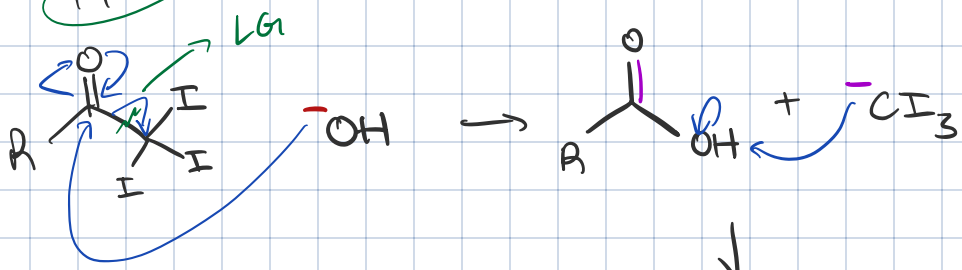


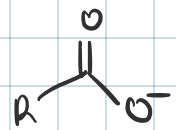
2.



replace these with I the same way as 1.

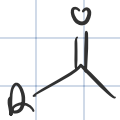
3.





$+ \; CHI_3 (s)$

What tests positive



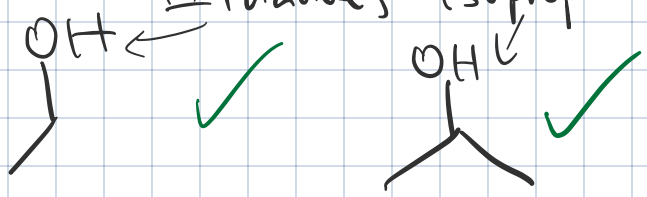
Methyl Ketones ✓

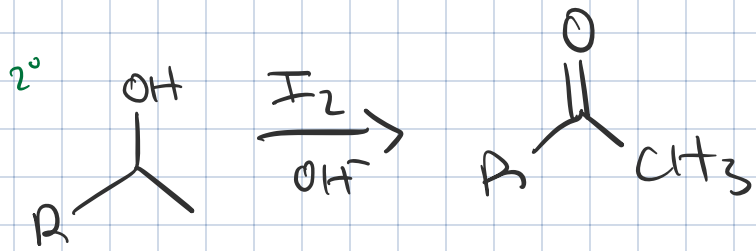
Acetaldehyde ✓

Alcohols that can be oxidized into the previous 2 by I_2

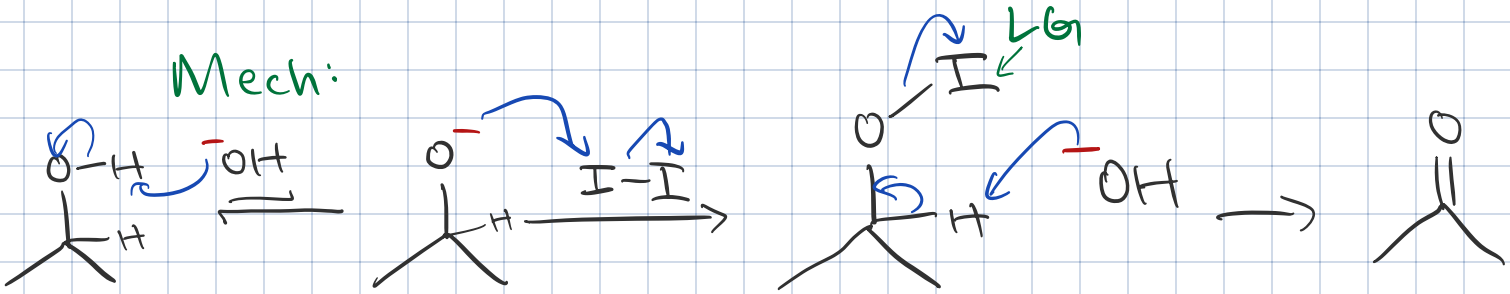
→ 2° methyl alcohols + Ethanal

Ethanol, Isopropanol



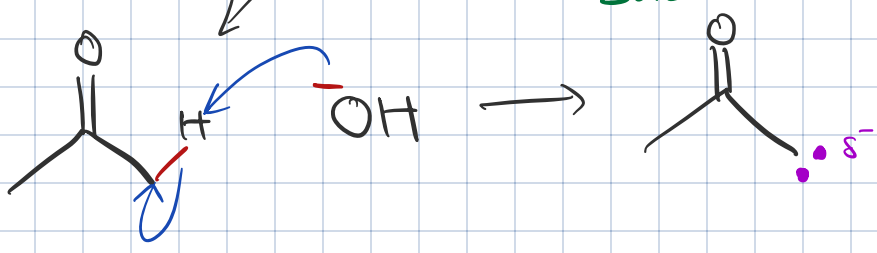


Mech:

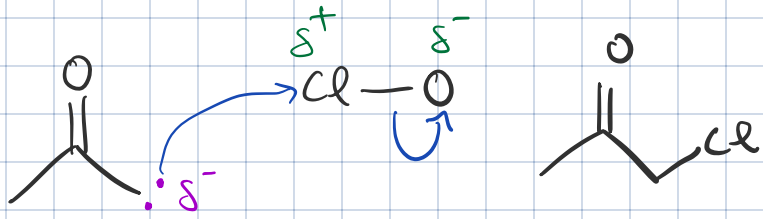


Common Haloform Rxn: Acetone + Bleach → Chloroform + Acetate

Basic

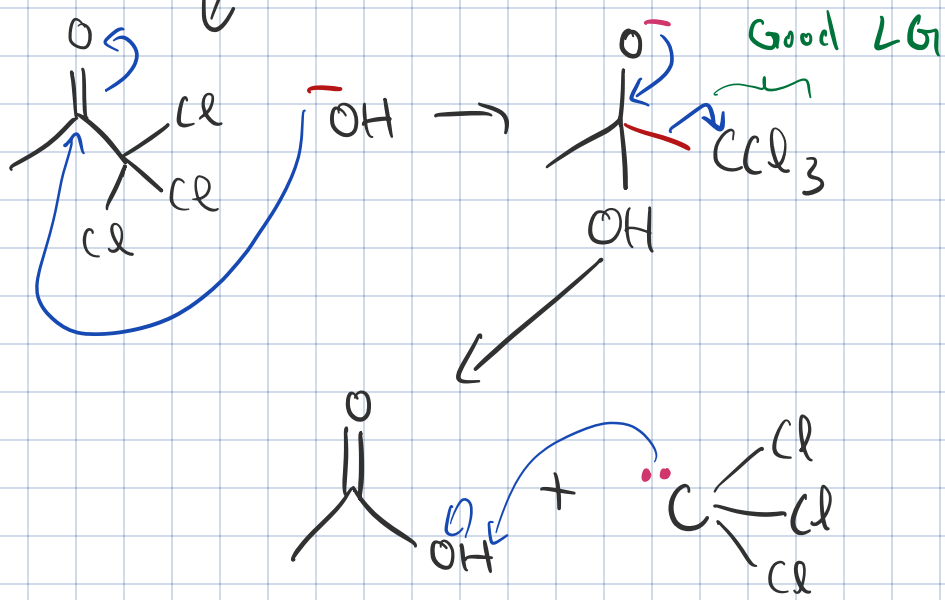


↔ enolate



Important Note: Cl inductive effect pushes successive α-chlorination to be geminal

repeat 3 times



$NaCH_3COO$ + $CHCl_3$